HEATSAFE PAKISTAN

Four-city heat-index ML pilots | Nationwide weather monitoring

HACKINSIGHT 2026 / OCTOBER 21, 2026 / ONE-PAGE PROJECT SUMMARY

4
EVALUATED ML CITIES

16
CITIES IN WEATHER FEED

210,432
HISTORICAL WEATHER HOURS

21
PREFECT RUN RECORDS

PROBLEM AND INTENDED USERS

Outdoor construction workers, supervisors and safety officers need understandable heat information to plan precautions. HeatSafe combines weather evidence, historically evaluated 24-hour heat-index prediction, uncertainty and cited safety advice; it does not certify any worksite as safe.

DATA PROVENANCE AND DATA QUALITY

ERA5 via Open-Meteo (2020-2025): 52,608 hourly weather observations per city for Karachi, Lahore, Islamabad and Multan (210,432 total). Historical CAMS modelled air-quality records cover Karachi only (26,304 hours, 2023-2025). Source manifests record URL, checksum and licence. Karachi has raw/cleaned/processed/gold DuckDB tables and rule-level fix/fill/flag logs. New-city weather datasets passed basic hourly completeness checks; full DQ-agent processing remains pending.

HISTORICAL ML EVALUATION: TARGET = HEAT INDEX 24 HOURS AHEAD

Separate HistGradientBoosting models were assessed against the same-hour previous-day persistence baseline. Chronological split: training 2020-2023; validation Jan-Jun 2024; interval calibration Jul-Dec 2024; held-out test 2025 (8,760 hours per city). Lower MAE is better.

CITY	MODEL MAE	BASELINE MAE	90% INTERVAL COVERAGE
Karachi	1.056 C	1.107 C	95.38%
Lahore	1.875 C	1.948 C	95.37%
Islamabad	1.798 C	1.852 C	93.32%
Multan	1.638 C	1.713 C	95.03%

All four models beat the baseline on the historical 2025 test set. Test risk-category accuracy: Karachi 87.41%, Lahore 81.06%, Islamabad 80.90%, Multan 85.42%. Coverage values are observed historical coverage for nominal 90% intervals; they do not guarantee live performance.

AI ASSISTANCE, AUTOMATION AND LIVE SCOPE

The Insight RAG Agent retrieves official CDC NIOSH, UK HSE and NOAA guidance; Gemini produces cited explanations and refuses certain unsupported requests. Numerical Karachi ML forecasts are read from verified DuckDB gold tables rather than invented by Gemini. Prefect completed auto-scheduled Karachi hourly runs at 2, 3, 4 and 5 PM on Oct 21; the evidence export contains 21 run records. A separate provider-issued 48-hour forecast feed covers 16 cities (768 records). Newly trained Lahore, Islamabad and Multan models are historical evaluations only; their live ML inference has not yet been integrated.

PRACTICAL VALUE, BUSINESS MODEL AND LIMITS

Potential users and payers include construction firms, site safety teams and public-interest pilots; possible revenue is employer subscriptions, subject to field validation and operating costs. Heat index is not measured WBGT, direct-sun exposure or a medical outcome. ERA5 is reanalysis, not an archive of forecasts issued at the time. The national weather feed is provider-generated, and the website/Hugging Face integration must be verified before final submission.

EVIDENCE modeling/evaluations/ | logs/ | evidence/prefect_runs.csv | manifest.csv

github.com/Hamdanilyas01/Hack_insight- | Verify final deployment status before submission